

Vinoth M

☎ +91 93603 79929 ✉ gopir525@gmail.com LinkedIn GitHub Portfolio

Experience

Associate Software Engineer

Sep 2024 – Present

Aptean

Coimbatore, India

- Defined HLD/LLD solution designs, data flow specs, and XML/WSDL/SOAP & REST integration standards for 15+ carrier APIs, including Amazon Logistics and FedEx on a logistics platform serving 50K+ daily requests with 99.9% uptime, collaborating with leads and architects.
- Architected a production RAG system using Azure OpenAI and pgvector semantic vector search, serving 8K queries/day at p99 latency of 450ms, reducing engineer lookup time by 53%.
- Engineered multi-agent AI workflows (Azure OpenAI) with Chain-of-Thought and Few-Shot prompting to parse specs, generate mapping configs, and trigger validation, cutting carrier onboarding effort by 70% (2 weeks → 3 days).
- Designed a metadata-driven carrier mapping engine applying SOLID principles and Factory and Strategy design patterns, enabling zero-code carrier onboarding and eliminating 40 hrs/month of manual configuration.
- Implemented OAuth 2.0 client credentials flow with JWT auth and RBAC across 14+ integrations; added Redis caching for carrier code lookups, cutting DB hits by 38% and reducing mapping-workflow p99 latency from 900ms to 350ms.
- Developed Java backend services with JDK 17+ (records, virtual threads, pattern matching) and multithreading for concurrent request handling; optimized React frontend via code-splitting, improving Time-to-Interactive by 40% for 5K daily active users.
- Maintained GitHub Actions CI/CD pipelines and deployed Node.js services on Azure App Services (Key Vault, Blob Storage); used the AWS SDK for CloudWatch/Elasticsearch monitoring, resolving 23+ production incidents and cutting average resolution time by 30%.
- Drove code reviews and resolved SonarQube static-analysis issues across a 10+ member team; coordinated Agile delivery with distributed teams spanning India, China, and the US; delivered sprint demos to stakeholders.

Projects

LinkedIn Content Agent | *Node.js, Gemini API, Multi-Agent AI, React, OAuth 2.0*

- Built a full-stack AI platform with a 3-agent pipeline (Planner, Writer, Reviewer) using Chain-of-Thought prompting to turn a topic into a publish-ready LinkedIn post; integrated OAuth 2.0 authorization code flow with token lifecycle management and a React UI featuring live agent reasoning traces and approval gates.

JobNest – Microservices Job Portal | *React.js, Redux, Spring Boot, MySQL, Redis, JWT*

- Engineered a job portal with Spring Boot microservices (auth, jobs, applications) secured with JWT and RBAC; designed a normalized MySQL schema with indexed queries reducing search latency by 35%; built a React + Redux frontend with role-based views, lazy loading, and client-side caching.

DocReady – Document Analysis & Migration Readiness Tool | *Python, Gemini AI, React, AWS SQS*

- Built a full-stack tool analyzing DOCX and PDF documents for knowledge-base migration readiness; used AWS SQS to asynchronously queue analysis jobs for concurrent processing, with an LLM-powered verdict engine producing structured readiness scores and actionable improvement suggestions.

Education

Sri Krishna College of Engineering and Technology

Dec 2021 – May 2025

B.Tech in Information Technology

Coimbatore, Tamil Nadu

- CGPA: 8.0 / 10.00
- Activities: Organized hackathons and codeathons, led department technical events, and mentored junior students in technical community initiatives.

Technical Skills

Languages: Java 17+, JavaScript, TypeScript, Python, SQL, C#

Backend / Frontend: Spring Boot, RESTful APIs, SOAP/WSDL, Microservices, Maven, Node.js, Express.js, React.js, Redux, .NET Core

AI / LLM: Azure OpenAI, RAG Systems, Multi-Agent AI, Prompt Engineering (CoT, Few-Shot), pgvector, Vector Search

Databases & Cloud: PostgreSQL, MySQL, SQL Server, Redis, AWS SDK, API Gateway, Lambda, ECS, EC2, RDS, S3, SQS, CloudWatch, Azure (App Services, Key Vault, Blob Storage), Docker, GitHub Actions, Elasticsearch

Messaging / Streaming: Apache Kafka, AWS Kinesis

Concepts & Tools: System Design (HLD/LLD), Data Structures and Algorithms, SOLID, Design Patterns, Multithreading, SonarQube, OAuth 2.0, OIDC, JUnit 5, JWT, Agile, Postman, Git

Achievements and Certifications

Oracle Database @ AWS Certified Architect Professional (2025) | AWS Academy Cloud Foundations | Red Hat OpenShift Applications ReactJS & Redux – Udemy | Java SE 8 – Infosys | Python – Coursera
Smart India Hackathon 2023 – qualified internal round (150+ teams) | LeetCode – 400+ problems solved